Fairness and Bias in Intelligent Interactive Systems - Evaluating Fairness and Bias in Multistakeholder Recommender Systems

1. Introduction

Recommender systems are a core component of contemporary social platforms, shaping users' interaction with content, products, information and each other. They receive a lot of attention from researchers aiming to develop frameworks and algorithms to guarantee fairness and eliminate bias from the item recommendation. However, many of the mitigation solutions' developers place emphasis on increasing "recommendation quality", often framing it as a one-dimensional optimization problem. Numerous experiments propose only offline solutions, which complement an analytical framework but may not stand the test of real user interaction with various interests that might not always align.

In a multi-stakeholder environment, which most real-world systems are, there are several parties present such as client users, providers and platform moderators with various definitions of "fairness" that often clash [1]. While fairness is a widely-used term in the computer science field, it originates from sociology, which means that bias and resulting conflicts of interests between different stakeholders may not always be a technical flaw, but an inherent feature of a multi-side market [22]. In that context evaluating bias and fairness in recommender systems requires moving from a single-metric analysis towards a meta perspective that considers various perspectives and long-term system maintainability [2, 17].

Therefore, the objective of this review is to define the common groups in which bias exists along with evaluation metrics, explore the relations between clashing stakeholders' objectives and resulting fairness optimization problems.

2. Research questions

2.1. Which categories of bias can be identified in contemporary recommender systems when classified by their place in the interaction pipeline and affected stakeholders?

2.2. What evaluation metrics are used for measuring fairness and performance in bias mitigation research, and which trade-offs emerge between competing objectives?

3. Search strategy

The study follows Systematic Literature Review methodology to capture the current state of technological research in recommender bias and fairness and lay groundwork for advancing knowledge by synthesizing relevant studies that may be useful for guiding future research.

To maintain quality and eligibility of findings, ACM digital library was selected as the main data source. Corresponding to the research questions posed and relevant topics in the same field, numerous search queries were formed and used to get a wider selection of papers. Only works published since February 2023 were considered to give an accurate representation of the status quo. The final queries are presented below:

- Title: ["Recommender system" OR "Recommendation"] AND ["Bias" OR "Fairness" OR "Bias mitigation" OR "Fairness Constraint" OR "Fair ranking"] AND "Multi-stakeholder"
- Title: ["Recommender" OR "Recommendation"] AND ["Bias" OR "Fairness" OR "Bias mitigation" OR "Fairness Constraint" OR "Fair ranking"] AND ["Intelligent" OR "AI" OR "Artificial Intelligence" OR "Diffusion" OR "Machine learning" OR "ML" OR "LLM" OR "GPT"]
- Title: "Recommender system" AND ["User control" OR "Interactive"]

- Title: [“Recommender system” OR “Recommendation”] AND [“Ranking system” OR “Discrimination”]
- Publication date: 01/02/2023 - 28/02/2026

These searches combined allowed retrieving 53 papers. Manual title and abstract screening eliminated 16, leading to 37 studies being saved in Mendeley reference manager for closer consideration. The protocol is shown in Figure 1. Full-text eligibility was then evaluated during careful reading, sorting out studies based on the following exclusion criteria:

- Overly short papers (Under 3 pages)
- Too niche and not representing overall bias and fairness trends in recommendation systems
- Focus on non-fairness related forms of bias (such as joint human-ai decision making bias)
- Not relevant in item recommendation context
- Low-novelty papers either describing incomplete research without proper intermediate result evaluation or providing an overgeneralised evaluation study with no new insights
- Insight on both bias evaluation and stakeholder impact absent

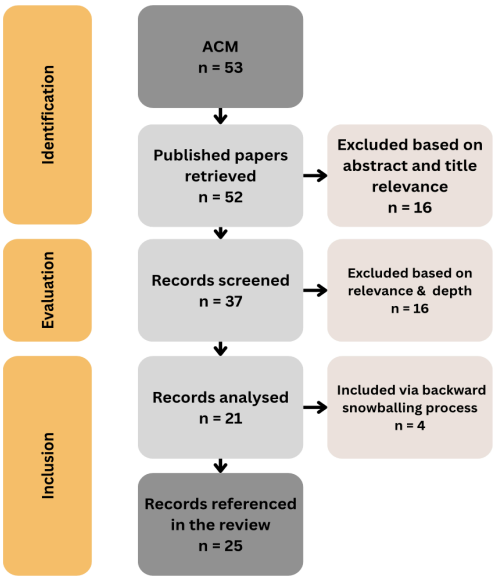

Figure 1: Study selection protocol

After rejecting 9 papers and putting 7 aside in the “backup” folder, the remaining 21 works were meticulously examined and classified using a tagging framework to identify main topics and patterns to review in the synthesis section, as seen in Table 1. 10 papers were nominated as anchor studies. Novelty was used as a filter based on tag uniqueness and diversity. This was followed by backward snowballing, adding 4 studies and resulting in 25 total references.

Table 1: Anchor studies coverage of bias type and mitigation objectives

Author	**Paper type**	**Bias types**	**Metrics**	**Affected parties**	**Miscellaneous**
Chen et al. [2]	Survey	Conformity, confounding, demographic, exposure, feedback loop, popularity, propensity, selection biases	Demographic and exposure parity, calibration quality, propensity score	Multi-stakeholder	Causal methods, future work, trade-offs, inter-stakeholder clash, fairness vs accuracy, positive bias exploitation
Peng et al. [17]	Algorithm	Representation bias	Interdependent metric based on diversity, accuracy and exposure parity	Multi-stakeholder	New metric
Zhang et al. [24]	Algorithm	Exposure, propensity biases, feedback loop	Accuracy parity	User	-
Krause et al. [11]	Evaluation study	Echo chamber, exposure and popularity biases, feedback loop	Exposure parity, long-tail coverage	Multi-stakeholder	Trade-offs
Ruan et al. [18]	Algorithm	Feedback loop	Accuracy parity	User	Rewriting biased training data with LLM
Malitesta	Evaluation	Demographic, exposure,	Accuracy parity,	Multi-stak	Trade-offs, user vs provider clash

et al. [15]	study	representation biases	exposure parity, long tail coverage	eholder	
Hou et al. [9]	Algorithm, framework	Exposure, popularity, representation bias, feedback loop	Accuracy parity, exposure parity	Multi-stak eholder	Trade-offs, user vs provider clash
Deldjoo [4]	Evaluation study	Exposure, recency, representation biases	Diversity, exposure parity	Provider	Design-stage bias prevention, user-controlled, trade-offs
Wang et al. [22]	Algorithm, evaluation study	Demographic bias, feedback loop	Calibration	User	Experiment
Gao et al. [7]	Evaluation study, framework	Echo chamber, filter bubble, feedback loop, exposure bias	Diversity, user satisfaction	User	Causal methods, dynamic reinforcement-based mitigation

4. Bias Categories in Recommender Systems (RQ1)

Bias in recommender systems may emerge at different stages of the interaction pipeline - from the data collection [6, 18] and ranking process to long-term feedback loops [7]. While mitigation strategies described in referenced works majorly include algorithm tuning to work on in-processing debiasing or post-factum reranking for most of the wide-spread bias types, a fundamental distinction concerns which stakeholder is disadvantaged, and how the bias is structurally produced and measured (refer to Table 1). These correlations were identified and studied in the synthesis section, resulting in the following bias families:

4.1. Interaction Data Biases

Interaction data biases mostly arise from either imbalanced content interaction and feedback on the user side, or the recommender system mischaracterising user-content interaction. Many real-world datasets feature power discrepancy where a small subset of items accumulates disproportionate attention.

Prominent cases like selection bias and exposure bias have items trapped due to distorted interaction data, since users cannot interact with content they do not see [11]. Other cases go even further and redistribute user attention based on metadata that is either irrelevant, like item's position in propensity bias [24], or even unknown like hidden variables in the case of confounding bias [25].

Although these biases still provide engagement, and may even go unnoticed on the user's side, they systematically **disadvantage content providers** who lose out on visibility due to circumstances beyond their control. While this group of biases is algorithmic, it has its origin in the users' interaction with the system.

4.2. Sociodemographic and Representation Bias

Demographic bias and representation bias arise from underrepresentation in the training data, model assumptions and recorded behavioural data. Wang et al. show that normally fairness violations from this category originate from outside the system and enter the technical scope in the form of datasets compromised by stereotypes and assumptions. As a consequence, misaligned models may fail to guarantee equal recommendation quality to various user groups, violate fairness and tolerance policies and reinforce social inequalities [15, 22].

On the contrary, this family of biases may also introduce individual unfairness to the system via misalignment bias, where a single user's taste differs from dominant trends [19].

Such biases raise concerns about both distributive and representational **fairness on the client's side** of the system. Unlike provider-focused bias, these mostly concern satisfaction with service, not exposure opportunities.

4.3. Dynamic Reinforcement

Often addressed as feedback loops, these biases describe a model's tendency to reinforce existing bias by considering its own output for future decisions. Seemingly successful recommendations influence user behaviour and allow a system to compromise its own interaction data that further shapes next recommendation iterations.

Over time, dynamic reinforcement may reduce informational diversity, favour dominant providers, and create echo chambers for users, as described by Gao et al. [7]. Noteworthy is their ability to effectively escalate any existing form of bias if left unchecked [2, 9, 11, 19, 22, 24].

While the effect is often unnoticed in the earlier stages and engagement rises, they narrow exposure spectrum over time, creating a phenomenon called filter bubble. Its temporal effect highlights the impossibility of evaluating a system's fairness at a singular point of time. A balanced recommender can drift towards reducing service satisfaction and performance **for all stakeholders involved.**

4.4. Interdependencies between categories

While the majority of papers referenced analytically distinct forms of bias, real world biases rarely function in isolation. Popularity and exposure biases, for instance, could be viewed both as an exposure imbalance and as a reinforcement phenomenon [7].

Similarly, as per Ruan et al., emotionally charged content, like negative news articles, often gets disproportionate engagement with the general population, due to social patterns, further blurring the line between categories [18].

Overall, biases are sociotechnological phenomena where static and dynamic components coexist. Data distribution, optimization strategies and user behaviour interact across time and stakeholders. Mitigation strategies that target one bias may inadvertently affect another. As a result, mitigation research overwhelmingly proposes case-specific interventions instead of universal solutions [2].

These interdependencies put emphasis on the multi-stakeholder perspective as fairness interventions reshape visibility and interaction distribution across actors and may create new forms of imbalance [23].

5. Evaluation metrics and Trade-offs (RQ2)

Interestingly, 19 out of the 25 mitigation studies reviewed feature the term "recommendation quality" as a metric for debiasing success [1-2, 4-9, 11-12, 14-19, 21-22, 24]. However, on a closer look, very few actually measure the same value in the experiments. There is no uniform definition of quality since stakeholders have wide-ranging interests and assume various priorities, with the most prominent examples listed below.

5.1. Commonly Used Evaluation Metrics

5.1.1. Accuracy-Oriented

Traditional recommender evaluation assumes metrics such as precision and recall, which measure how well recorded data of a user correlates with the recommendations they receive.

Conceptually, they prioritise individual relevance and chase short-term engagement. Among published algorithms this study covers, this metric family is prevalent [3, 5, 8, 18, 20, 21, 24].

5.1.2. Fairness-Oriented

Fairness metrics are values that assess disparity between system components in terms of fair interactions. Exposure parity prioritises equitable item visibility [4, 13], while user fairness metrics may assess performance between demographic groups or similar individuals [20].

Overall, such forms of evaluation consider distributional fairness as opposed to average performance - possibly interfering with accuracy objectives.

5.1.3. Stakeholder-Specific

Possible evaluation metrics are a very large set of solutions tailored for different stakeholders' needs. While users may appreciate diversity or accuracy, providers could covet short-term catalogue exposure or long-tail loyal audience. Platform metrics include stability or overall satisfaction. Instead of following standardized metrics, researchers can introduce new ones depending on their goals like Peng et al. [17].

Even for a singular stakeholder type different potential measurements of success and quality exist, which frames fairness as a relative metric [1, 2, 14].

5.2. Tradeoff analysis

Aird debates the perception of fairness as a single dimension task and reviews popular metrics such as fairness from the perspective of different actors in the system. Evidently, tuning recommendation algorithms almost always leads to clashes which require compromises and trade-offs between competing objectives [1].

5.2.1. Accuracy vs Exposure

Improving exposure for various content providers normally implies dethroning highly-relevant items and replacing them with less frequently interacted ones, which interferes with individual recommendations' accuracy, especially in the short-term. This clash is a conflict between different stakeholders' satisfaction that requires precise balance [11, 15, 23]. As noticed by Aird [1], consumers are often viewed as a protected group that takes precedence over providers.

5.2.2. Accuracy vs Fairness

Even across the same stakeholder group, interests may not align [1, 2]. Mitigating demographic bias and injustice is an inherently social issue. Attempting to implement in-processing technological solutions means changing the model's weights or introducing stricter constraints [14]. Recently, practitioners like Ruan et al. have been exploring the idea of utilising LLMs for pre-processing training datasets [18]. All aforementioned solutions alter the model's interpretation of the training data, potentially sacrificing accuracy.

5.2.3. Personalisation vs Diversity

Recommendation systems present an interesting consequence of overfitting - deep personalization means recommending very similar content and potentially trapping a user in an echo chamber, as seen in the work of Krause et al. [11] and Sguerra et al. [19] - reduced diversity is often noticed in the long-term, leading to dissatisfaction with the platform. Nevertheless, Li et al. illustrate solutions that manage to elevate performance on both metrics [12].

5.2.4. Bias Mitigation vs Bias Leverage

Chen et al. argue that a bias does not carry inherent harm to the system. For example, sociodemographic bias is in itself a necessary trade-off for the model to represent the human thinking process effectively, while popularity bias reflects item quality as one of its properties, which may increase performance if used well [2]. Various forms of visibility bias are irreplaceable in zero-shot scenarios [4]. Considering the described clashes and many more that come up when mitigating bias, a thorough understanding and overview take precedence over complete elimination of bias.

6. A win-win scenario?

In certain contexts, fairness and performance objectives align in their goal to improve every stakeholder's satisfaction of the platform. Increased diversity influences long-term user satisfaction, which may often lead to larger interaction volume for the providers. Merinov and Ricci [16] explore frameworks for positive-sum impact in multistakeholder-systems by modelling the users'

limited knowledge to both optimise their utility and promote underexposed items. However, interventions mean redistribution of utility and visibility and over-tuning might lead to disproportionate satisfaction, which is why a delicate balance is crucial [9, 10].

7. Conclusions

In conclusion, the reviewed literature reveals three distinct yet interconnected bias families. Crucially, they do not operate in isolation, as seen with popularity bias which functions as an exposure imbalance and a reinforcement issue simultaneously. Hence, bias mitigation strategies in recommender systems are not a one-dimensional optimization problem and careless implementation of these methods introduces potential drawbacks, possibly compromising the system [16]. The included studies highlight how the majority of fairness enhancements are a trade-off which redistributes unfairness to a different factor, possibly changing the disadvantaged stakeholder [4, 14, 15, 23]. Despite the advancements in recommendation algorithms, not every single one can be implemented without proper review [1].

Reviewed studies employ a wide, yet fragmented set of evaluation metrics. Although 19 of the 25 studies frame their goal as improving "recommendation quality," no two studies define that term identically, confirming that quality is stakeholder-relative. While researchers and developers aim simply to improve overall satisfaction, proper planning at design time is necessary to avoid greater harm to an unexpected participant as shown in described trade-off scenarios. Positive-sum scenarios are possible [10, 16], but remain context-dependent and require accurate calibration to avoid redistributing disadvantage [9, 23]. Considering the abundance of available metrics [2], strategy developers should always form a very clear research goal at design time and understand its alignment and system-level impact. This literature review can serve as an introduction to the multi-objective problem of stakeholder-aware fairness balance in recommender system frameworks.

Possible limitations include reliance on the ACM digital library as the only source of research and the 3-year publication window. Future research should focus on standardised multi-stakeholder fairness benchmarks and account for potential discrepancy between offline fairness evaluation and real user interaction.

8. References

1. Aird, A. (2025). Addressing Multi-stakeholder Fairness Concerns in Recommender Systems Through Social Choice. *RecSys2025 - Proceedings of the 19th ACM Conference on Recommender Systems*, 1439–1444. https://doi.org/10.1145/3705328.3748756
2. Chen, J., Dong, H., Wang, X., Feng, F., Wang, M., & He, X. (2023). Bias and Debias in Recommender System: A Survey and Future Directions. *ACM Transactions on Information Systems*, *41*(3). https://doi.org/10.1145/3564284
3. Chen, Q., Wei, B., Cheng, D., Li, J., Liu, L., & Zhang, S. (2024). A Novel Shadow Variable Catcher for Addressing Selection Bias in Recommendation Systems. *Proceedings - IEEE International Conference on Data Mining, ICDM*, 71–80. https://doi.org/10.1109/ICDM59182.2024.00014
4. Deldjoo, Y. (2026). Understanding Biases in ChatGPT-based Recommender Systems: Provider Fairness, Temporal Stability, and Recency. *ACM Transactions on Recommender Systems*, *4*(2), 1–35. https://doi.org/10.1145/3690655
5. Deng, J., Chen, Q., Cheng, D., Du, X., Li, J., & Liu, L. (2025). Mitigating Latent Confounding Bias in Recommender Systems. *CIKM 2025 - Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, 531–541. https://doi.org/10.1145/3746252.3761177
6. Ferrara, C., Casillo, F., Gravino, C., de Lucia, A., & Palomba, F. (2024). ReFAIR: Toward a Context-Aware Recommender for Fairness Requirements Engineering. *Proceedings - International Conference on Software Engineering*, 2631–2642. https://doi.org/10.1145/3597503.3639185
7. Gao, C., Wang, S., Li, S., Chen, J., He, X., Lei, W., Li, B., Zhang, Y., & Jiang, P. (2023). CIRS: Bursting Filter Bubbles by Counterfactual Interactive Recommender System. *ACM Transactions on Information Systems*, *42*(1). https://doi.org/10.1145/3594871
8. Hong, J., Jeon, G., & Lee, S. (2023, December 6). Exploring User-oriented Social Recommendation System through Granting Users Control over a Social Group. *Proceedings of the 5th ACM*

International Conference on Multimedia in Asia, MMAsia 2023. https://doi.org/10.1145/3595916.3626369

9. Hou, Y., Kou, Y., Shen, D., Zhou, X., Li, D., Nie, T., & Yu, G. (2025). LeadFairRec: LLM-enhanced Discriminative Counterfactual Debiasing for Two-sided Fairness in Recommendation. *CIKM 2025 - Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, 867–876. https://doi.org/10.1145/3746252.3761126
10. Hu, Y., Lyu, Z., Bai, L., & Cui, L. (2025). FairWork: A Generic Framework For Evaluating Fairness In LLM-Based Job Recommender System. *SIGIR 2025 - Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 3964–3968. https://doi.org/10.1145/3726302.3730145
11. Krause, T., Deriyeva, A., Beinke, J. H., Bartels, G. Y., & Thomas, O. (2025). Mitigating Exposure Bias in Recommender Systems—A Comparative Analysis of Discrete Choice Models. *ACM Transactions on Recommender Systems*, *3*(2), 1–37. https://doi.org/10.1145/3641291
12. Li, H., Shen, D., Wang, C., Liu, Y., & Gu, J. (2025). Can LLMs Enhance Fairness in Recommendation Systems? A Data Augmentation Approach. *SIGIR 2025 - Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 570–580. https://doi.org/10.1145/3726302.3729917
13. Li, H., Wu, K., Zheng, C., Xiao, Y., Wang, H., Geng, Z., Feng, F., He, X., & Wu, P. (2023). *Removing Hidden Confounding in Recommendation: A Unified Multi-Task Learning Approach*.
14. Liu, Z., Xu, X., Yu, J., Xu, H., Hu, L., Li, H., & Gai, K. (2024). A Self-Adaptive Fairness Constraint Framework for Industrial Recommender System. *International Conference on Information and Knowledge Management, Proceedings*, 4726–4733. https://doi.org/10.1145/3627673.3680099
15. Malitesta, D., Medda, G., Purificato, E., Marras, M., Malliaros, F., & Boratto, L. (2025). How Fair is Your Diffusion Recommender Model? *RecSys2025 - Proceedings of the 19th ACM Conference on Recommender Systems*, 1199–1205. https://doi.org/10.1145/3705328.3759318
16. Merinov, P., & Ricci, F. (2024). Positive-Sum Impact of Multistakeholder Recommender Systems for Urban Tourism Promotion and User Utility. *RecSys 2024 - Proceedings of the 18th ACM Conference on Recommender Systems*, 939–944. https://doi.org/10.1145/3640457.3688173
17. Peng, Y., Qian, X., & Song, W. (2023). A Re-ranking Approach for Two-sided Fairness on Recommendation Systems. *ACM International Conference Proceeding Series*, 312–316. https://doi.org/10.1145/3603781.3603836
18. Ruan, Q., Xu, J., Leavy, S., mac Namee, B., & Dong, R. (2024). Rewriting Bias: Mitigating Media Bias in News Recommender Systems through Automated Rewriting. *UMAP 2024 - Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization*, 67–77. https://doi.org/10.1145/3627043.3659541
19. Sguerra, B., Epure, E. v., Lee, H., & Moussallam, M. (2025). Biases in LLM-Generated Musical Taste Profiles for Recommendation. *RecSys2025 - Proceedings of the 19th ACM Conference on Recommender Systems*, 527–532. https://doi.org/10.1145/3705328.3748030
20. Souza, R., & Manzato, M. (2024). A Two-Stage Calibration Approach for Mitigating Bias and Fairness in Recommender Systems. *Proceedings of the ACM Symposium on Applied Computing*, 1659–1661. https://doi.org/10.1145/3605098.3636092
21. Tommasel, A. (2024). Fairness Matters: A look at LLM-generated group recommendations. *RecSys 2024 - Proceedings of the 18th ACM Conference on Recommender Systems*, 993–998. https://doi.org/10.1145/3640457.3688182
22. Wang, C., Wang, K., Bian A., Islam, R., Keya, K. N., Foulds, J., & Pan, S. (2023). When Biased Humans Meet Debiased AI: A Case Study in College Major Recommendation. *ACM Transactions on Interactive Intelligent Systems*, *13*(3). https://doi.org/10.1145/3611313
23. Xu, C., Chen, S., Xu, J., Shen, W., Zhang, X., Wang, G., & Dong, Z. (2023). P-MMF: Provider Max-min Fairness Re-ranking in Recommender System. *ACM Web Conference 2023 - Proceedings of the World Wide Web Conference, WWW 2023*, 3701–3711. https://doi.org/10.1145/3543507.3583296
24. Zhang, G., Yuan, G., Cheng, D., Liu, L., Li, J., & Zhang, S. (2025). Mitigating Propensity Bias of Large Language Models for Recommender Systems. *ACM Transactions on Information Systems*, *43*(6). https://doi.org/10.1145/3736404
25. Zhang, Q., Zhang, X., Liu, Y., Wang, H., Gao, M., Zhang, J., & Guo, R. (2023). Debiasing Recommendation by Learning Identifiable Latent Confounders. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 3353–3363. https://doi.org/10.1145/3580305.3599296